

WASP-12b

R-0371

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230305 · created 2026-07-18T16:51:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460009.7321 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.072 +/- 0.03
Transit depth (Rp/Rs)^2	0.52 +/- 0.43 [%]
Orbital Inclination (inc)	78.87 +/- 2.83
Airmass coefficient 1 (a1)	1.009 +/- 0.023
Airmass coefficient 2 (a2)	-0.0073 +/- 0.0084
Scatter in the residuals of the lightcurve fit is	2.29 %
Best Comparison Star	#1 - [371, 263]
Optimal Method	PSF photometry
Transit Duration (day)	0.104 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R ² fit vs published	0.072 ± 0.03 vs 0.1178 (deviation 1.35σ)
Duration fit vs published	0.104 d vs 0.125 d (deviation 1.55σ)
Mid-time O–C	1573.3 min
Depth SNR	2.1
Residual scatter	2.29 %

Light curve

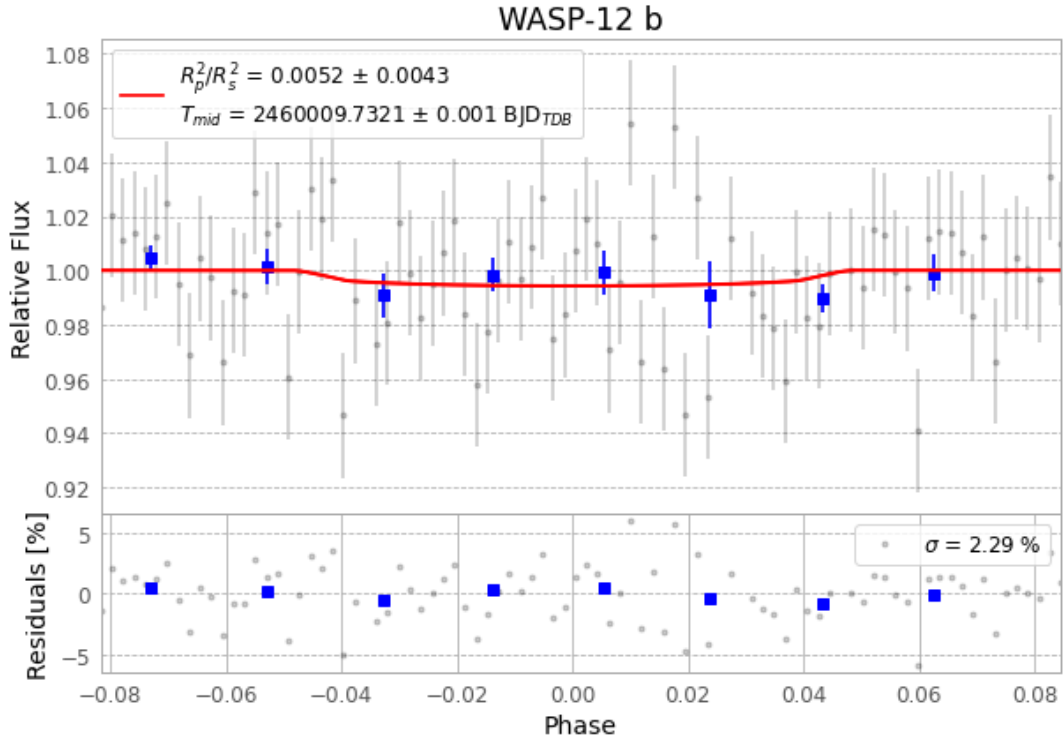


Figure 1 — FinalLightCurve_WASP-12 b_2023-03-05.png (SHA-256 ae62c6ec8fa4c0dc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d91bdef49053fbc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 69abe98

Data provenance

85 FITS frames (56 MB), WASP-12230306032115.FITS → WASP-12230306074215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230305.log (fdf9a989c180...), FinalLightCurve_WASP-12 b_2023-03-05.png (ae62c6ec8fa4...), FinalLightCurve_WASP-12 b_2023-03-05.csv (b4b8710c2ae9...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 16:51Z.